Besides *PgHero*, there are commercial tools like PgAnalyze that cam help you identify missing indexes. PgAnalyze offers a free tool called *PgAnalyze Index Advisor*.[12]

Besides manually adding indexes, Index creation can even be automated using some advanced tools. The HypoPG[13] extension and dexter[14] tool can automatically create indexes.

You've now seen some ways to build up your query observability tool set.

While PGSS is a critical tool, one limitation is that it shows you a normalized query. Once you've identified a specific poor query group with PGSS, you will want to gather a sample from production. From the sample of a real execution of a query in that group, you can look at the EXPLAIN plan to see whether there's a missing Index or something else to improve.

Later, you'll keep diving further into query analysis. But right now a piece in the query observability tool chain is missing.

From PGSS, you'll want to collect real query samples. Once equipped with those samples, you'll be able to dive into their details. Let's see a couple of methods to get those samples.

Logging Slow Queries

PGSS is a critical tool, but you'll need sample queries within a group to analyze further.

One way is to log slow queries into postgresql.log. You can do this by setting the log_min_duration_statement parameter to a value like 1 second, to log any query that takes longer. From psql run SHOW data_directory; to find the data directory path. In this directory you'll find your postgresql.conf file and the log directory where your log file is located. Run SELECT pg_current_logfile(); to see the current log file.

In the config file, set the following parameter and save the file. Reload your config with pg_ctl reload --pgdata=DATADIR, replacing DATADIR with the value from SHOW data_directory;.

```
log_min_duration = 1000
```

12. https://pganalyze.com/index-advisor
13. https://github.com/HypoPG/hypopg
14. https://github.com/ankane/dexter